

2023

JOURNEYMAN

OPEN

BOOK

EXAM #12



25 QUESTIONS

TIME LIMIT - 1 HOUR

TIME SPENT **MINUTES**

SCORE **%**

JOURNEYMAN OPEN BOOK EXAM #12 25 Questions One Hour Time Limit

1. If an exit enclosure (stair tower) is required to be separated from the building, only electrical wiring methods serving equipment permitted by the ____ is permitted within the exit enclosure.

(a) electrical inspector (b) building inspector (c) fire marshal (d) AHJ

2. For large-scale PV installations (5,000 kW or more), buildings whose sole purpose is to house supply station equipment shall ____ comply with 690.12 for rapid shutdown systems.

(a) never (b) be required to (c) not be required to (d) always

3. The emergency shutoff device for a fuel dispenser must simultaneously disconnect all conductors of the circuit(s), including the ____ conductor if any.

(a) grounded (b) grounding electrode (c) equipment grounding (d) all of these

4. Unless otherwise permitted, wiring for ____ loads shall be kept independent from all other wiring and equipment.

(a) emergency (b) legally-required standby (c) optional standby (d) all of these

5. A building may have more than one service if the capacity requirement exceeds ____ amperes at 1,000 volts or less.

(a) 1,000 (b) 1,200 (c) 2,000 (d) 2,400

6. Equipment intended to interrupt current at fault levels shall have a/an ____ rating, at normal voltage, that is at least equal to the available fault current at the line terminals of the equipment.

(a) time delay (b) interrupting (c) short-time (d) long-time

7. Microgrid systems shall be permitted to disconnect from other sources and operate in ____.

(a) stand by (b) stand alone (c) parallel (d) island mode

8. Openings around electrical penetrations into or through fire-resistance rated ____ shall be firestopped using approved methods.

(a) walls (b) floors (c) ceilings (d) all of these

9. The dc conductors of a PV system shall not occupy the same _____ as PV system ac conductors unless separated by a barrier or partition.

- (a) raceway (b) cable (c) enclosure (d) all of these**

10. On a 4-wire, delta-connected system where the midpoint of one phase winding is grounded, the conductor having the higher voltage to ground shall be marked by an outer finish that is orange or _____.

- (a) gray (b) yellow (c) brown (d) other effective means**

11. The voltage at the load terminals of a fire pump controller shall not drop more than _____ percent below the voltage rating of the connected motor when operating at 115% of the full-load current of the motor.

- (a) 1.2 (b) 3 (c) 5 (d) 7.5**

12. Vegetation such as trees shall **not** be used for support of _____.

- (a) luminaires (b) boxes (c) overhead conductor spans (d) any of these**

13. A/an _____ listed shall be installed between a wind-electric system and any loads served by the premises wiring system.

- (a) surge protective device (b) GFCI (c) AFCI (d) arrester**

14. A luminaire in a commercial cooking hood must, among other requirements, exclude grease, oil, and _____ from the lamp and wiring compartment.

- (a) solids (b) moisture (c) oxygen (d) exhaust vapors**

15 A device that uses power electronics to convert one form of electrical power into another form of electrical power is known as a/an _____.

- (a) solar cell (b) wind turbine (c) inverter (d) electronic power converter**

16. Stainless steel rigid metal conduit may use galvanized steel boxes and enclosures if those enclosures are not subject to _____.

- (a) severe corrosive influences (b) unauthorized contact**
(c) corrosive influences (d) physical damage

17. For a PV system, the rapid shutdown for conductors inside the array boundry must limit the voltage to not more than ____ volts within 30 seconds of the shutdown initiation.

- (a) 10 (b) 15 (c) 24 (d) 30**

18. The sum of the multiconductor cable fill area as a percentage of the allowable fill area for the tray calculated in accordance with 392.22(A), and the single-conductor cable fill area as a percentage of the allowable fill area for the tray calculated in accordance with 392.22(B), totals not more than ____%.

- (a) 75 (b) 80 (c) 100 (d) 125**

19. The maximum voltage of an ESS shall be the rated ESS input and ____ voltage(s).

- (a) output (b) line (c) secondary (d) load**

20. There shall be no voltage marking on a Type TC cable employing ____ wire.

- (a) covered (b) fixture (c) rubber insulated (d) thermocouple extension**

21. The maximum size FMT permitted is ____.

- (a) 1/2" (b) 3/4" (c) 1" (d) 1 1/4"**

22. Fastening of unbroken lengths of EMT conduit can be increased to a distance of ____ from the termination point where the structural members do not readily permit fastening within 3 feet.

- (a) 4' (b) 5' (c) 6' (d) 10'**

23. The minimum size copper equipment grounding conductor required on a motor branch circuit with a 30 amp circuit breaker and #12 copper conductor is ____.

- (a) #10 (b) #8 (c) #12 (d) #14**

24. What is the maximum length of an unprotected feeder tap conductor?

- (a) 10 feet (b) 25 feet (c) 50 feet (d) 100 feet**

25. Individual open conductors and cables other than service entrance cables shall not be installed less than ____ feet from grade level.

- (a) 8 (b) 10 (c) 15 (d) 18**

